```
import random

machines = ['A', 'B', 'C']
N = [0, 0, 0]
R = [[], [], []]
Q = [0, 0, 0]

epsilon = 0.2

rewards_pattern = {
    0: [3, 4, 2, 5],
    1: [6, 7, 5, 8],
    2: [4, 3, 5, 4]
}
rewards_idx = [0, 0, 0]
```

```
for i in range(10):

    r = random.random()

    if r < epsilon:
        x = random.randint(0, 2)
    else:
        max_q = max(Q)
        best_machines_indices = [idx for idx, q_val in enumerate(Q) if q_val == max_q]
        x = random.choice(best_machines_indices)

    print("\nPlay", i + 1)
    print("Machine selected:", machines[x])

    if rewards_idx[x] < len(rewards_pattern[x]):
        reward = rewards_pattern[x][rewards_idx[x]]
        rewards_idx[x] += 1
        print("Reward:", reward)
    else:
        print("Warning: Reward pattern for machine", machines[x], "exhausted. Assigning 0 reward.")
        reward = 0

    N[x] = N[x] + 1
    R[x].append(reward)
    Q[x] = Q[x] + (reward - Q[x]) / N[x]
```

```
Play 1
Machine selected: A
Reward: 3

Play 2
Machine selected: A
Reward: 4

Play 3
Machine selected: A
Reward: 2

Play 4
Machine selected: A
Reward: 5

Play 5
Machine selected: A
Warning: Reward pattern for machine A exhausted. Assigning 0 reward.

Play 6
Machine selected: B
Reward: 6

Play 7
Machine selected: A
Warning: Reward pattern for machine A exhausted. Assigning 0 reward.

Play 8
Machine selected: B
Reward: 7

Play 9
Machine selected: B
Reward: 5

Play 10
Machine selected: B
Reward: 8
```

```
print("\nFinal Results")
print("Machine  N  Rewards  Q")

for i in range(3):
    print(machines[i],N[i],R[i],round(Q[i],2))
```

```
Final Results
Machine  N  Rewards  Q
A 6 [3, 4, 2, 5, 0, 0] 2.33
B 4 [6, 7, 5, 8] 6.5
C 0 [] 0
```